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Data Structures and Algorithms Project Report

Topic: SORTING ALGORITHMS

Course: Data Structures & Algorithms

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1 Group Information

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**Note: All members participated in Time Calculator implementation, Report writing, Citations, and Final checking.*

2 Algorithms Presentation

2.1 Selection Sort [3]

2.1.1 Overview

- **History and Origin:** A simple and intuitive in-place comparison-based sorting algorithm.
- **Efficiency:** Generally performs worse than similar algorithms like Insertion Sort. Best suited for situations where memory writes are costly.
- **Application:** Useful for small datasets or when auxiliary memory is extremely limited.

2.1.2 Idea

- The algorithm divides the input list into two parts: a sorted sublist built up from left to right at the front, and a sublist of the remaining unsorted items.
- Initially, the sorted sublist is empty and the unsorted sublist is the entire list.
- It proceeds by finding the smallest element in the unsorted sublist, swapping it with the leftmost unsorted element, and moving the sublist boundary one element to the right.

2.1.3 Step-by-Step Algorithm

- Consider an array of n elements, a[0. . . (n - 1)].
- **Step 1:** Start a loop with variable i from 0 to n - 2.
- **Step 2:** Assume the element at i is the minimum ($min = i$).
- **Step 3:** Iterate through the remaining array from $j = i + 1$ to n - 1 to find the actual minimum element.
  - If a[j] ¡ a[min], update $min = j$.
- **Step 4:** Swap a[i] with a[min].
- **Step 5:** Increase i by 1 and repeat until the array is fully sorted.

2.1.4 Pseudo Code

- **Algorithm:** Selection Sort
- **Input:** An array A of n elements

```
i = 0
While i < n - 1:
   min = i
   For j = i + 1 to n - 1:
      If A[min] > A[j]:
         min = j
   Swap(A[i], A[min])
   i = i + 1
```

2.1.5 Complexity

- **Time Complexity:**
  - Best case: $O(n^2)$
  - Average case: $O(n^2)$
  - Worst case: $O(n^2)$

    The algorithm scans the entire remaining array regardless of the initial order.
- **Space Complexity:** $O(1)$

  In-place algorithm, requiring only a constant amount of extra memory for swapping.

2.1.6 Variants and Optimizations

- **Current Implementation (Standard):**
  - The provided C++ code implements the standard unidirectional Selection Sort, which scans for the minimum element from left to right in a single pass.
- **Theoretical Variant: Bidirectional Selection Sort [3]:**
  - **Problem:** Conventional Selection Sort only finds the minimum element in each pass, leaving the rest of the array unsorted.
  - **Solution:** In each pass, simultaneously find both the minimum and maximum elements, placing them at the beginning and end of the unsorted section respectively.
  - **Benefit:** Reduces the total number of iterations by half compared to the implemented version, improving practical execution time.

2.2 Insertion Sort [2]

2.2.1 Overview

- **History and Origin:** One of the oldest and most intuitive sorting algorithms. Mimicking how humans sort a hand of playing cards.
- **Efficiency:** Best for small datasets or nearly sorted/sorted data. Worst for reverse-sorted datasets.
- **Application:** Often used as the base case for hybrid algorithms by Tim Sort, Intro Sort, Quick Sort, etc. to handle small sub-lists.

2.2.2 Core Concepts

- Let the list be divided by sub-lists, separating them into sorted and unsorted sections.
- Take the first element of the unsorted list and insert it into its proper place in the sorted one.
- After each placement, the size of the sorted region grows by 1 and the size of the unsorted region shrinks by 1.

2.2.3 Step-by-Step Explanation

- Consider an array of n elements, $a[0 \dots (n-1)]$.
- **Step 1:** Start an increase variable i that is from 1 to $n-1$.
- **Step 2:** Find the correct position pos in $a[0 \dots (i-1)]$ to insert $a[i]$, where $a[pos-1] < a[i] < a[pos]$.
  - Set $val = a[i]$, then move forward $a[pos \dots (i-1)]$ to one element. After that, set $a[pos] = val$.
  - Increase i by 1.
- **Step 3:** If $i < n$ return to **Step 2**. Otherwise, stop the algorithm.

2.2.4 Pseudo Code

- **Algorithm:** Insertion Sort
- **Input:** An array A of n elements

```
For i = 1 to n - 1:
    val = A[i]
    j = i - 1
    While j >= 0 and A[j] > key:
        A[j + 1] = A[j]
        j = j - 1
    A[j + 1] = key
```

2.2.5 Complexity Analysis

- **Time Complexity:**
  - Best case: $O(n)$
    When the input is already sorted.
  - Average case: $O(n^2)$
    When the input is random.
  - Worst case: $O(n^2)$
    When the input is reverse-sorted.
- **Space Complexity:** O(1)
  In-place algorithm. Requires only a constant amount of extra memory.

2.2.6 Variants and Optimizations

- **The "Hit" Method (Enhanced Insertion Sort)[10]:**
  - **Problem:** Conventional Insertion Sort is inefficient for reverse-sorted lists due to maximum comparisons.
  - **Solution:** Immediately compare $A[i]$ with $A[0]$ (the current minimum).
  - **Benefit:** If $A[i] < A[0]$, it is moved to the front instantly, bypassing all intermediate comparisons and significantly reducing execution time.
- **Binary Insertion Sort:**
  - Use Binary Search to find the position of an element within the sorted sub-array which is greater than or equal to A[i].
  - While Binary Search reduces the number of comparisons to $O(n \log n)$, the complexity of data movement remains $O(n^2)$ because elements must still be shifted to make space for the new entry.

2.3 Shell Sort [1]

2.3.1 Overview

- **Origin:** The shell sort is a popular variant of insertion sort. It is first published in 1959 by Donald Shell.
- **Efficiency:** This algorithm is appropriate for medium-sized and sorted/nearly sorted datasets.
- **Application:** The shell sort can be applied in reality to deal with medium-sized data to replace other sorting algorithms namely Merge sort, which are too heavy. Sometimes, this technique can be used as a subroutine in hybrid algorithms.

2.3.2 Core concepts

- The shell sort uses the same operating principles with insertion sort. However, the key difference is that Shell sort instead of comparing adjacent elements to find place for the insertion, it compares elements which are far apart.

2.3.3 Step-by-step Expanation

- Assume that the array of n elements, a[0...(n-1)].
- **Step 1:** Set the gap base on a certain sequence , usually $(n/2)$
- **Step 2:** Divide the array into several sublist and do insertion sort in those.
- **Step 3:** Reduce the gap and repeat the process on **Step 2** if the gap is greater than 1.
- **Step 4:** If the gap equals to 1, do the last insertion sort when the array is nearly sorted.

2.3.4 Pseudo code

```
gap = n/2
while gap >= 1:
    for i = gap to n-1:
        key = a[i]
        j = i - gap
        while (j >= 0 and a[j] > key):
            a[j + gap] = a[j]
            j = j - gap
        a[j + gap] = a[j]
        gap = gap / 2
```

2.3.5 Complexity Analysis

- Best case: $O(nlogn)$

  If choose the suitable sequence for the array.
- Worst case: $O(1.5n)$

  If choose the wrong sequence for the problem.
- Average case: $O(n^{1.25})$

2.3.6 Variants and Optimizations

- Variants of Shell sort is mainly the choose of sequence:
  - Knuth's Increments: $1, 3, 13, ..., (3k - 1)/2$
  - Sedgewick's Increments: $1, 8, 23, 77, ...$
  - Hibbard's Increments: $1, 3, 7, 15, 31, ...$
- Each Optimization has its own pros and cons. Depend on the input data, we can choose a distinct version to achieve best performance

2.4 Bubble Sort [5]

2.4.1 Overview

- **Origin:** The bubble sort probably had its origins in a paper published in 1956 by Friend entitled "Sorting on Electronic Computer Systems," which uses the term "sorting by exchange"
- **Efficiency:** This algorithm is only used for small datasets.
- **Application:** The bubble sort can be used for educational purposes because it is one of the simplest sorting algorithms to study.

2.4.2 Core concepts

- The bubble sort compares adjacent elements and swaps their position if neccessary.
- The algorithm works to swap the larger items to the end of the array, those items are described as bubbles float to the surface, and that is the reason this sorting algorithm is called bubble sort.

2.4.3 Step-by-step Expanation

- Assume that the array of n elements, a[0...(n-1)].
- **Step 1:** Start with the first element of the array.
- **Step 2:** Compare each pair of adjacent elements and swap them if they are in the wrong order.
- **Step 3:** After the algorithm compares the last pair of elements, the largest element is moved to the end of the array
- **Step 4:** Repeat the process at **Step 2** and **Step 3** with the remaining unsorted part.
- The algorithm will stop after (n-1) passes .

2.4.4 Pseudo code

```
1. for i = 0 to n - 1:
    2.   for j = 0 to n - i - 1:
        3.      if a[j] > a[j+1]:
            4.      swap(a[j], a[j+1])
```

2.4.5 Complexity Analysis

For original version of bubble sort:

- Best case: $O(n^2)$

  When the array is already sorted, it requires 0 swap but the number of comparisons remain unchanged.

- Worst case: $O(n^2)$

  When the array is in reversed order, the number of swaps is maximum

- Average case: $O(n^2)$

2.4.6 Variants and Optimizations

- **Current implementation:** Original Bubble
  - Compares all adjacent pair of elements.
- **Improved implementation** Bubble sort with additional flag:
  - The flag is bool type to check whether the array is sorted or not.
    - If the array is already sorted, the algorithm will stop immediately
    - This enhancement reduces time complexity of bubble sort to $O(n)$ in best case instead of $O(n^2)$ in all cases.

2.5 Heap Sort [11]

2.5.1 Overview

- **Origin:** Heapsort was invented by J. W. J. Williams in 1964.
- **Efficiency:** Heap sort is a highly efficient comparison-based sorting algorithm that utilizes a binary heap structure.
- **Application:** Heap sort can be applied to large datasets due to its consistent running time of $O(nlogn)$ for all cases.

2.5.2 Core Concepts

- Heap sort is an improved version of Selection sort, which operates based on the principle of building a Max-Heap from unsorted elements to sort arrays in ascending order.
- The root of the heap is always the largest element and it will be placed in its first correct position in the sorted portion.

2.5.3 Step-by-step Explanation

- Assume an array of n elements: $a[0 \dots (n-1)]$.
- The algorithm consists of two main stages:
- **Stage 1: Heap construction**
  - Construct a binary heap from elements indexed from 0 to $n-1$.
  - Rearrange elements so that the largest element becomes the root of the heap (the element at index 0).
- **Stage 2: Sorting**
  - Swap the root element with the last element in the unsorted portion.
  - Reduce the heap size by 1 and repeat the process at step 2 with the new heap size
- The algorithm terminates when all elements are sorted.

2.5.4 Pseudo code

```
Heapify(A, start, n)
 leftChild = start * 2 + 1
 rightChild = start * 2 + 2
 LargerChild = leftChild
 if leftChild >= n: return
 if A[leftChild] < A[rightChild]:
     LargerChild = rightChild
 if A[start] < A[LargerChild]:
     swap(A[start], A[LargerChild])
      Heapify(A,LargerChild,n)
```

```
BuildMaxHeap(A,n)
for i = n-1 down to 1:
    swap(A[0], A[i])
    heapSize = heapSize - 1
    Heapify(A, 0, heapSize)
```

2.5.5 Complexity Analysis

- **Best case:** $O(n \log n)$
- **Average case:** $O(n \log n)$
- **Worst case:** $O(n \log n)$

2.5.6 Variants and Optimizations

- Variants of this algorithm is the adjustment for the number children.
- **Current implementation:** Binary heap: Build Max-Heap with each node has at most two children (Left and Right)
- **Another implementation:** Ternary heap [6]: each node has at most three children

2.6 Merge Sort [4]

2.6.1 Overview

- **History and Origin:** Merge Sort is a divide-and-conquer algorithm that was invented by John von Neumann in 1945. Merge sorting is also known as a divide and conquer method of sorting features, and it is based on this method (Varshney & Yadav, 2014).
- **Efficiency:** Consistently performs well on large datasets.
- **Application:** Commonly used in external sorting and when stable sorting is required.

2.6.2 Core Concepts

- The array is recursively divided into two halves until each sub-array contains a single element.
- The sub-arrays are then merged back together in a sorted manner.

2.6.3 Step-by-Step Explanation

- Consider an array of n elements, $a[0 \ldots (n-1)]$.
- **Step 1:** Recursively divide the array into halves until it reaches sub-arrays of size 1.
- **Step 2:** Merge the sorted sub-arrays back together by using two pointers to track the current position in the left and right sub-arrays, respectively. If there are remaining elements in the left sub-array, copy them to a temporary array. Repeat the same for the right sub-array.
- **Step 3:** Copy the sorted temporary array back to the original array $a[left \ldots right]$.
- **Step 4:** Repeat the process until the entire array is merged and the recursion stack is empty.

2.6.4 Pseudo Code

- **Algorithm:** Merge Sort
- **Input:** An array A of n elements

```
MergeSort(arr[], l, r):
    If l < r:
        middle index: m = l + (r - l)/2
        Call mergeSort(arr, l, m)
        Call mergeSort(arr, m + 1, r)
        Call merge(arr[], l, m, r) to merge the two halves

merge(arr[], l, m, r):
    Set pos = left, leftPos = left, rightPos = mid + 1
    While leftPos <= mid and rightPos <= right:
        tempArray[pos++] = min(a[leftPos], a[rightPos])
        If a[leftPos] <= a[rightPos]: increment leftPos
        else: increment rightPos
```

2.6.5 Complexity Analysis

- **Time Complexity:**
  - Best case: $O(n \log n)$
  - Average case: $O(n \log n)$
  - Worst case: $O(n \log n)$
- **Space Complexity:** O(n)

  Requires additional memory for the temporary array used in the merge process.

2.6.6 Variants and Optimizations

- **The Hybrid Sort (Mixing between Quick Sort and Merge Sort)**
  - **Step 1:** Split the array [a] into [n] parts.
  - **Step 2:** Select part [1]
  - **Step 3:** Sort it using Quick Sort.
  - **Step 4:** Select part [2] and implement the same process until part [n] is sorted.
  - **Step 5:** Merge the sorted parts using Merge Sort.

2.7 Quick Sort [7]

2.7.1 Overview

- **History and Origin:** Developed by Tony Hoare in 1959. One of the most famous divide-and-conquer algorithms.
- **Efficiency:** Highly efficient and generally outperforms other $O(n \log n)$ algorithms in practice due to excellent cache locality.
- **Application:** The default sorting algorithm in many standard libraries (e.g., C++ `std::sort`).

2.7.2 Idea

- It works by selecting a 'pivot' element from the array and partitioning the other elements into two sub-arrays according to whether they are less than or greater than the pivot.
- The sub-arrays are then sorted recursively.
- The base cases are empty arrays or arrays of a small size, where a simpler algorithm (like Insertion Sort) is often used to finish sorting.

2.7.3 Step-by-Step Algorithm

- **Step 1:** If the array segment has less than 10 elements, apply **Insertion Sort** to quickly sort it and terminate the recursion.
- **Step 2:** Select a pivot using the **Median-of-Three** method (finding the median of the first, middle, and last elements).
- **Step 3:** Swap the chosen pivot with the second-to-last element to temporarily move it out of the way.
- **Step 4:** Use two pointers, *left* and *right*, starting from the ends of the segment.
  - Move *left* forward until finding an element larger than the pivot.
  - Move *right* backward until finding an element smaller than the pivot.
  - Swap these two elements if $left \leq right$.
- **Step 5:** Restore the pivot to its correct final position.
- **Step 6:** Recursively apply Quick Sort to the sub-array before the pivot and the sub-array after the pivot.

2.7.4 Pseudo Code

- **Algorithm:** Quick Sort (Optimized)
- **Input:** An array A, left index l, right index r

```
1. If (r - l) < 10:
2.    InsertionSort(A, l, r)
3.    Return
4. p = Partition(A, l, r) // Uses Median-of-Three
```

```
5. QuickSort(A, l, p - 1)
6. QuickSort(A, p + 1, r)
```

2.7.5 Complexity

- **Time Complexity:**
  - Best case: $O(n \log n)$
  - Average case: $O(n \log n)$
  - Worst case: $O(n^2)$

    Occurs when the partition is highly unbalanced, though the Median-of-Three optimization makes this extremely rare.
- **Space Complexity:** $O(\log n)$

  Bounded by the call stack depth during recursion.

2.7.6 Variants and Optimizations

- **Implemented Optimization 1: Median-of-Three Pivot Selection**
  - **Implementation:** The `selectPivot` function in the provided C++ code chooses the median of the first, middle, and last elements $(A[l], A[mid], A[r])$ to use as the pivot.
  - **Benefit:** Avoids the $O(n^2)$ worst-case performance on already sorted or reverse-sorted arrays, ensuring a more balanced partition in practical scenarios.
- **Implemented Optimization 2: Hybrid Sorting (Insertion Sort Threshold)**
  - **Implementation:** The `quickSortArray` function checks if the sub-array size is less than 10 (`(r - l) < 10`). If so, it halts recursion and delegates the sorting to `insertionSortSubmatrix`.
  - **Benefit:** Eliminates the overhead of recursive function calls for very small arrays, where simple $O(n^2)$ algorithms like Insertion Sort execute much faster due to lower constant factors.

2.8 Radix Sort [9]

2.8.1 Overview

- **History and Origin:** A non-comparative sorting algorithm originally used for sorting punch cards.
- **Efficiency:** Highly efficient for datasets with keys that are not too large (e.g., integers or strings).
- **Application:** Widely used in environments where numerical data needs to be sorted in linear time relative to the number of digits.

2.8.2 Idea

- Radix Sort avoids direct comparison between elements. Instead, it groups elements by their individual digits.
- The implemented version is Least Significant Digit (LSD) Radix Sort, which processes digits from the least significant (rightmost) to the most significant (leftmost).
- It relies on a stable sorting subroutine (like Counting Sort) to sort the digits at each positional place value.

2.8.3 Step-by-Step Algorithm

- Consider an array of n elements.
- **Step 1:** Find the maximum value ($maxv$) in the array to determine the number of digits.
- **Step 2:** Initialize a multiplier $j = 1$.
- **Step 3:** While $maxv$ / j ¿ 0, perform Counting Sort for the current digit place:
  - Create a frequency array $Count$ of size 10 (for base 10) initialized to 0.
  - Calculate occurrences of each digit: $digit$ = (array[i] / j) % 10.
  - Accumulate $Count$ array to determine actual positions.
  - Place elements into a temporary array B traversing backwards to maintain stability.
  - Copy array B back to the original array.
- **Step 4:** Multiply j by 10 and return to **Step 3** until all digits are processed.

2.8.4 Pseudo Code

- **Algorithm:** LSD Radix Sort
- **Input:** An array A of n elements

```
1. Find maxv in A
2. j = 1
3. While (maxv / j) > 0:
4.    Count = array of 10 zeros
5.    For i = 0 to n - 1:
```

```
        digit = (A[i] / j) % 10
        Count[digit]++
    For k = 1 to 9:
        Count[k] = Count[k] + Count[k - 1]
    B = array of size n
    For i = n - 1 down to 0:
        digit = (A[i] / j) % 10
        B[Count[digit] - 1] = A[i]
        Count[digit]--
    Copy B back to A
    j = j * 10
```

2.8.5 Complexity

- **Time Complexity:**
  - Best, Average, and Worst case: $O(d(n+k))$

    Where d is the number of digits, n is the number of elements, and k is the base (10).
- **Space Complexity:** $O(n+k)$

  Requires auxiliary arrays (*Count* and B) for the internal Counting Sort.

2.8.6 Variants and Optimizations

- **Current Implementation (Base-10 LSD):**
  - The provided C++ code implements Least Significant Digit (LSD) Radix Sort using base 10 (decimal system). It extracts digits using modulo (`% 10`) and allocates a counting array of size 10 for each pass.
- **Theoretical Variant: Bitwise Radix Sort (Base-256):**
  - **Problem:** Using base 10 requires division (`/`) and modulo (`%`) operations, which are computationally expensive. Additionally, base 10 requires more passes for large integer limits.
  - **Solution:** Change the radix base to a power of 2 (e.g., base 256 for byte-level extraction) using fast bitwise operations (right shift `>>` and bitwise AND `&`).
  - **Benefit:** Drastically reduces the number of sorting passes (e.g., only 4 passes for 32-bit integers) and replaces slow arithmetic operations with fast bitwise logic.

2.9 Binary Insertion Sort

2.9.1 Overview

- **History and Origin:** Binary insertion sort is a well-known variant of insertion sort that uses binary search to determine the correct insertion position. This technique was mentioned by John Mauchly in 1946 in one of the earliest published discussions on computer sorting.
- **Efficiency:** Best for small datasets or nearly sorted/sorted data. Worst for reverse-sorted datasets.
- **Application:** Instead of using insertion sort when the subarray size becomes small in hybrid algorithms such as Timsort, Introsort, or Quicksort, binary insertion sort can be used for better efficiency. It is also useful in problems where the cost of comparisons is relatively high.

2.9.2 Idea

- Let the list be divided by sub-lists, separating them into sorted and unsorted sections.
- Take the first element of the unsorted list and insert it into its proper place in the sorted one.
- After each placement, the size of the sorted region grows by 1 and the size of the unsorted region shrinks by 1.

2.9.3 Step-by-Step Algorithm

- Binary insertion sort uses the same code structure as Insertion Sort, except for the process of finding the correct position. Therefore, only this step is described below.
- Find the correct position *pos* in $a[0 \ldots (i-1)]$ to insert a[i], where a[pos - 1] $\leq$ a[i] ¡ a[pos].
- **Step 1:** Set val = a[i], $L = 0$, R = i - 1 and *pos* = i.
- **Step 2:** While L $\leq$ R.
  - Set Mid = (L + R) / 2.
  - If a[Mid] $\geq$ val then set R = Mid - 1 and pos = Mid.
  - Else set L = Mid + 1.
- **Step 3:** Move forward a[pos ... (i - 1)] to one element. After that, set a[pos] = val.

2.9.4 Pseudo Code

- **Algorithm:** Binary Insertion Sort
- **Input:** An array A of n elements

```
for i ← 1 to length(A)-1
     value ← A[i]
     l ← 0, r ← i-1, pos ← i
     while l <= r
```

```
        m ← l + (r-l) / 2
        if A[m] >= value
            pos ← m
            r ← m-1
        else
            l ← m+1
    for j ← i downto pos+1
        A[j] ← A[j-1]
    A[pos] ← value
```

2.9.5 Complexity

- **Time Complexity:**
  - Finding correct position takes $O(logn)$ but the process of insertion still takes $O(n)$.
  - Best case: $O(nlogn)$

    When the input is already sorted.
  - Average case: $O(n^2)$

    When the input is random.
  - Worst case: $O(n^2)$

    When the input is reverse-sorted.
- **Space Complexity:** $O(1)$

  In-place algorithm. Requires only a constant amount of extra memory.

2.9.6 Variants and Optimizations

- **Binary Insertion Sort:**
  - Use Binary Search to find the position of an element within the sorted sub-array which is greater than or equal to A[i].
  - While Binary Search reduces the number of comparisons to $O(n \log n)$, the complexity of data movement remains $O(n^2)$ because elements must still be shifted to make space for the new entry.

2.10 Shaker Sort

2.10.1 Overview

- **History and Origin:** One of the oldest and most intuitive sorting algorithms. Mimicking how humans sort a hand of playing cards.
- **Efficiency:** Best for small datasets or nearly sorted/sorted data. Worst for reverse-sorted datasets.
- **Application:** Often used as the base case for hybrid algorithms by Tim Sort, Intro Sort, Quick Sort, etc. to handle small sub-lists.

2.10.2 Idea

- Let the list be divided by sub-lists, separating them into sorted and unsorted sections.
- Take the first element of the unsorted list and insert it into its proper place in the sorted one.
- After each placement, the size of the sorted region grows by 1 and the size of the unsorted region shrinks by 1.

2.10.3 Step-by-Step Algorithm

- Consider an array of n elements, a[0. . . (n - 1)].
- **Step 1:** Start an increase variable i that is from 1 to n - 1.
- **Step 2:** Find the correct position *pos* in a[0. . . (i - 1)] to insert a[i], where a[pos - 1] ¡ a[i] ¡ a[pos].
  - Set val = a[i], then move forward a[pos. . . (i - 1)] to one element. After that, set a[pos] = val.
  - Increase i by 1.
- **Step 3:** If i ¡ n return to **Step 2**. Otherwise, stop the algorithm.

2.10.4 Pseudo Code

- **Algorithm:** Shaker Sort
- **Input:** An array A of n elements

```
l ← 0
r ← length(A) - 1
while l < r
     pos ← r
     for i ← l+1 to r
          if A[i] < A[i-1]
               swap A[i], A[i-1]
               pos ← i-1
     r ← pos
     for i ← r-1 downto l
```

```
        if A[i] > A[i+1]
            swap A[i], A[i+1]
            pos ← i+1
    l ← pos
```

2.10.5 Complexity

- **Time Complexity:**
  - Best case: $O(n)$
    When the input is already sorted.
  - Average case: $O(n^2)$
    When the input is random.
  - Worst case: $O(n^2)$
    When the input is reverse-sorted.
- **Space Complexity:** O(1)
  In-place algorithm. Requires only a constant amount of extra memory.

2.10.6 Variants and Optimizations

- **The "Hit" Method (Enhanced Insertion Sort)[10]:**
  - **Problem:** Conventional Insertion Sort is inefficient for reverse-sorted lists due to maximum comparisons.
  - **Solution:** Immediately compare $A[i]$ with $A[0]$ (the current minimum).
  - **Benefit:** If $A[i] < A[0]$, it is moved to the front instantly, bypassing all intermediate comparisons and significantly reducing execution time.
- **Binary Insertion Sort:**
  - Use Binary Search to find the position of an element within the sorted sub-array which is greater than or equal to A[i].
  - While Binary Search reduces the number of comparisons to $O(n \log n)$, the complexity of data movement remains $O(n^2)$ because elements must still be shifted to make space for the new entry.

2.11 Flash Sort[8]

2.11.1 Overview

- **History and Origin:** Invented by Karl-Dietrich Neubert in 1997.
- **Efficiency:** Generally effective for sorting large datasets with a uniform distribution of values, achieving linear time complexity in the best case.

2.11.2 Core Concepts

- FlashSort is based on the principle of *in-situ* permutation, combining the concepts of bucket sort and counting sort.
- The algorithm operates by classifying elements into different *classes* based on their values, then permuting them directly into their target positions.
- It achieves $O(n)$ time complexity under the assumption of a uniform distribution of input values.

2.11.3 Step-by-Step Explanation

- Consider an array of n elements, $a[0 \dots (n-1)]$.
- **Step 1:** Identify classes size
$$m = 0.45 \cdot n$$
(0.45 is considered by scientists, which gives the best performance for the algorithm) and create an auxiliary list L of classes to hold the number of elements in each one.
- **Step 2:** Each element $a[i]$ is classified into a class based on its value which is calculated using the formula:
$$k = \lfloor (m-1) \cdot \frac{a[i] - min}{max - min} \rfloor$$
where min and max are the minimum and maximum values in the array, respectively.
- **Step 3:** Convert the class counts L into a prefix sum, where L_b $(b \in \mathbb{Z}, 0 \leq b < m)$ is the number of elements $a[i]$ in class b and the previous classes $(L_{m-1} = n)$, which
$$L_b = \sum_{i=0}^{b} L_i$$
- **Step 4:** Rearrange the elements $a[i]$ in the array by permuting them into their correct classes when $L_{b-1} < i \leq L_b$.
- **Step 5:** Finally, applying Insertion Sort to sort the array.

2.11.4 Pseudo Code

- **Algorithm:** Insertion Sort
- **Input:** An array A of n elements

```
If n <= 1: return
m = floor(0.45 * n)
Initialize L[0...m-1] with 0

minVal = A[0], maxIndex = 0
For i = 1 to n - 1:
    If A[i] < minVal: minVal = A[i]
    If A[i] > A[maxIndex]: maxIndex = i

c1 = (m - 1) / (A[maxIndex] - minVal)
For i = 0 to n - 1:
    k = floor(c1 * (A[i] - minVal))
    L[k] = L[k] + 1
For i = 1 to m - 1:
    L[i] = L[i] + L[i - 1]

Swap(A[0], A[maxIndex])
move = 0, j = 0, k = m - 1
While move < n - 1:
    While j >= L[k]:
        j = j + 1
        k = floor(c1 * (A[j] - minVal))
    flash = A[j]
    While j != L[k]:
        k = floor(c1 * (flash - minVal))
        L[k] = L[k] - 1
        pos = L[k]
        Swap(flash, A[pos])
        move = move + 1

Call InsertionSort(A)
```

2.11.5 Complexity Analysis

- **Time Complexity:**
  - Best case: $O(n)$

    When the input is uniformly distributed, leading to balanced class sizes and minimal permutations.

    All elements are already in the correct class, resulting Insertion Sort being applied to very small sub-arrays.
  - Average case: $O(n)$
  - Worst case: $O(n^2)$

    Occurs when the input has some extremely large or small elements while most elements are the same, leading to unbalanced class sizes, causing like Insertion Sort to be applied to large sub-arrays.
- **Space Complexity:** O(m)

  Only requires additional space for the class count array L.

2.11.6 Variants and Optimizations

- **Change Insertion Sort to another sorting algorithm:**
  - If the input data is in the worst case, the algorithm performs Insertion Sort, resulting $O(n^2)$ time complexity.
  - Avoiding this by changing Insertion Sort to another sorting algorithm (e.g., Quick Sort, Merge Sort or Heap Sort), if the size of data is larger than a threhold.

3 Experimental results and comments

3.1 Running time of algorithms

The four line graphs below illustrate the running time of 12 sorting algorithms on four different input orders: sorted, nearly sorted, reverse sorted, and random.
Overall, it is clear that the figure for Selection Sort and Bubble Sort have significantly higher running times compared to the other algorithms.g

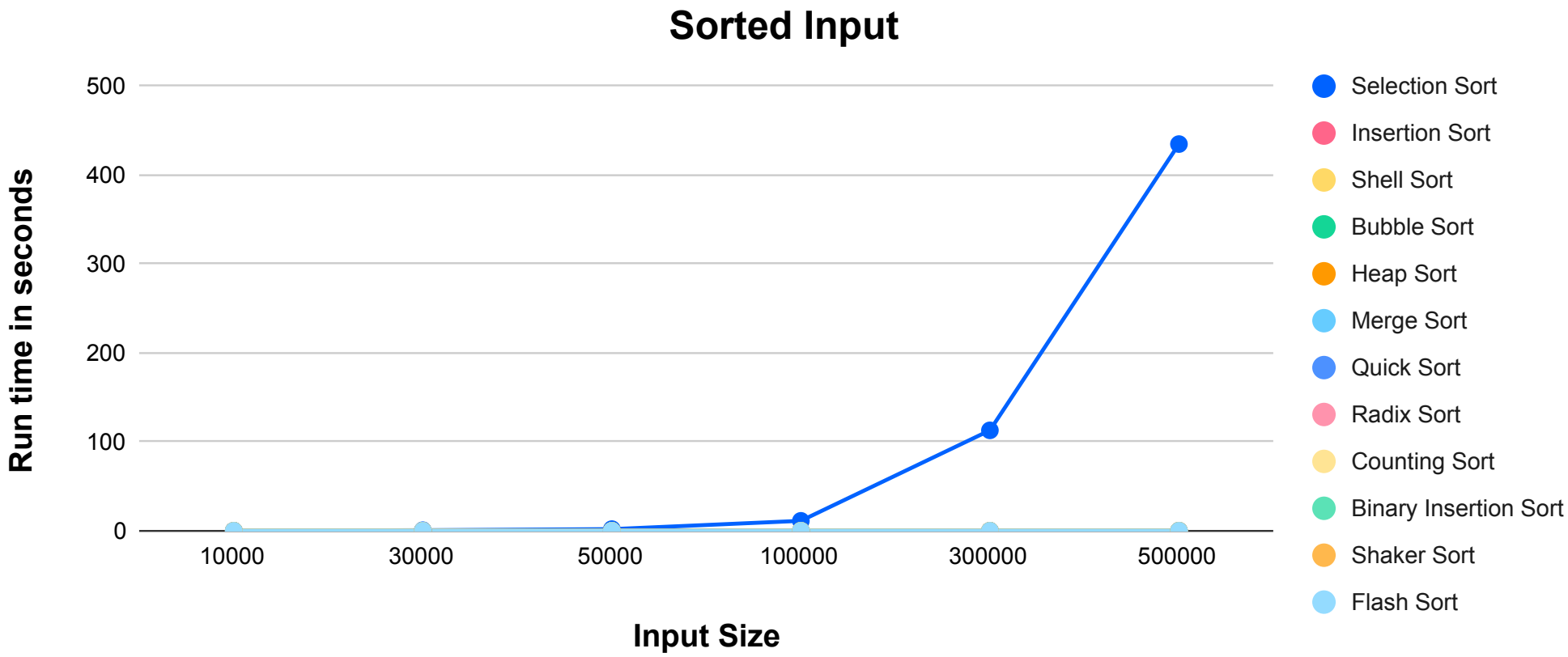
3.2 The number of comparisons in algorithms

Data order: **Sorted**												
Data size	10.000		30.000		50.000		100.000		300.000		500.000	
Resulting statics	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision
Selection Sort	105,167	100009999	689,787	900029999	1917,92	2500049999	11233,5	10000099999	112996	90000299999	434892	2,5E+11
Insertion Sort	0,0514	29998	0,0714	89998	0,7709	149998	3,5999	299998	1,3967	899998	6,0458	1499998
Shell Sort	0,6222	360042	2,8272	1170050	4,0681	2100049	8,5708	4500051	30,0859	15300061	49,7864	25500058
Bubble Sort	0,0295	19999	0,0883	59999	0,132	99999	0,2964	199999	0,8674	599999	3,6275	999999
Heap Sort	3,289	518705	13,5221	1739633	15,6552	3056481	46,292	6519813	155,882	21431637	199,605	37116275
Merge Sort	1,9953	465243	6,1066	3534740	12,507	11271929	24,5689	25699104	77,2574	62516397	142,383	153959690
Quick Sort	0,5135	133080	1,1606	462298	1,2239	824587	1,8735	1749164	5,8564	5498317	10,8987	9696622
Radix Sort	0,3049	90012	1,1599	330015	1,9931	550015	3,6059	1100015	13,9273	3900018	23,0738	6500018
Counting Sort	0,1188	70005	0,3401	210005	1,513	350005	2,2584	700005	3,4015	2100005	6,1422	3500005
Binary Insertion Sort	0,2556	287231	0,7632	954463	1,7852	1668927	5,3163	3537855	9,0504	11551423	25,3334	19951423
Shaker Sort	0,0668	40000	0,174	120000	0,2898	200000	0,579	400000	1,744	1200000	4,8341	2000000
Flash Sort	0,5451	118994	2,0147	356994	2,8529	594994	5,6932	1189994	24,029	3569994	38,9354	5949994

Data order: **Nearly Sorted**												
Data size	10.000		30.000		50.000		100.000		300.000		500.000	
Resulting statics	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision
Selection Sort	82,4528	100009999	682,385	900029999	3006,82	2500049999	13195,6	10000099999	141754	90000299999	429977	2,5E+11
Insertion Sort	0,3599	170474	0,4457	381370	6,1393	553362	2,9462	744174	2,0722	1227934	8,1505	1886330
Shell Sort	0,9578	360042	4,5803	1170050	4,9382	2100049	10,0758	4500051	31,5097	15300061	50,6101	25500058
Bubble Sort	145,838	19999	1264,74	59999	2640,79	99999	8307,18	199999	10576,3	599999	44942,5	999999
Heap Sort	2,9228	518705	14,3961	1739633	18,676	3056481	41,7754	6519813	132,22	21431637	183,995	37116275
Merge Sort	2,5839	964668	6,2076	5147796	12,0211	14036437	24,3352	31436444	76,5169	80950733	142,079	185575804
Quick Sort	1,077	133100	1,2535	462338	1,7467	824627	2,0375	1749204	5,581	5498361	11,6933	9696658
Radix Sort	0,3253	90012	1,1225	330015	2,1678	550015	3,622	1100015	14,3872	3900018	23,4548	6500018
Counting Sort	0,1108	70005	0,3532	210005	1,4122	350005	2,0138	700005	3,2722	2100005	5,7755	3500005
Binary Insertion Sort	0,4763	322535	2,1051	1180677	2,5163	1940779	13,9861	3836605	9,9932	11803373	26,5061	20076435
Shaker Sort	0,5658	151888	1,6826	441699	2,2423	620061	2,8785	821044	5,2285	1158616	3,3031	1368945
Flash Sort	0,694	118970	1,6874	356968	2,6621	594968	6,765	1189970	23,3154	3569970	30,9164	5949971

Data order: **Reverse Sorted**												
Data size	10.000		30.000		50.000		100.000		300.000		500.000	
Resulting statics	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision
Selection Sort	74,7506	100009999	680,337	900029999	1838,97	2500049999	17064,2	10000099999	153636	90000299999	425904	2,5E+11
Insertion Sort	112,237	100009999	988,149	900029999	6096,3	2500049999	27393,4	10000099999	226738	90000299999	6,06E+05	2,5E+11
Shell Sort	1,1906	360042	3,4599	1170050	6,4618	2100049	13,6358	4500051	44,2755	15300061	76,1319	25500058
Bubble Sort	443,437	19999	3969,7	59999	11402,4	99999	47834,7	199999	560380	599999	1,52E+06	999999
Heap Sort	2,7592	518705	8,7699	1739633	19,4824	3056481	46,6272	6519813	127,902	21431637	175,502	37116275
Merge Sort	1,8889	1431110	6,5497	6691262	11,0356	16720383	24,2663	37104342	90,0728	99359047	137,091	217412214
Quick Sort	0,4261	228055	3,779	805555	1,8849	1351331	3,3012	2902241	11,198	9807440	19,6311	17408348
Radix Sort	0,3199	90012	1,1688	330015	2,2355	550015	3,2913	1100015	13,7003	3900018	22,3289	6500018
Counting Sort	0,1023	70005	0,3383	210005	1,3279	350005	1,3663	700005	2,9557	2100005	4,9697	3500005
Binary Insertion Sort	72,2163	50252260	643,266	450849494	1781,94	1251493960	13474,8	5003187890	124015	45010501462	353376	1,2502E+11
Shaker Sort	466,945	100005001	4280,14	900015001	11441,7	2500025001	47519,8	10000050001	423172	90000150001	1186810	2,5E+11
Flash Sort	0,6865	103752	1,9107	311252	2,5955	518752	5,9086	1037502	25,004	3112502	32,9099	5187502

Data order: **Randomized**												
Data size	10.000		30.000		50.000		100.000		300.000		500.000	
Resulting statics	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision	Running time	Comparision
Selection Sort	72,8876	100009999	675,09	900029999	1878,82	2500049999	12825,3	10000099999	220180	90000299999	728805	2,5E+11
Insertion Sort	55,086	50098568	946,173	450995386	3474,48	1242358588	15996,1	5005123027	164705	44941469846	486114	1,2483E+11
Shell Sort	2,7062	360042	9,2453	1170050	15,0898	2100049	32,5809	4500051	109,411	15300061	200,162	25500058
Bubble Sort	394,146	19999	3819,66	59999	10771,3	99999	46402,3	199999	607584	599999	1,49E+06	999999
Heap Sort	5,033	517875	15,7985	1740722	28,8383	3059571	43,9528	6521229	142,695	21444835	260,311	36875028
Merge Sort	3,3413	2004825	14,3255	8599102	21,4229	20053445	43,5151	44170450	142,988	122441839	244,4	257294351
Quick Sort	0,9036	223823	4,7699	743646	4,1647	1304635	9,22	2738622	28,7321	8809663	44,3427	15272160
Radix Sort	0,3412	90012	1,1721	330015	2,1878	550015	3,3941	1100015	11,4715	3300015	20,378	5500015
Counting Sort	0,1238	69999	1,4104	210005	1,3746	315541	1,1992	565541	2,805	1565541	4,857	2565541
Binary Insertion Sort	36,3726	25194614	322,132	226583161	877,095	625351350	7217,7	2501105806	61707,3	22529618643	171356	6,2512E+10
Shaker Sort	307,156	67044246	2785,63	602121328	8012,19	1672044062	22268,3	6656689407	277446	60046607924	781174	1,6678E+11
Flash Sort	0,9527	96409	2,3163	288519	4,8677	466016	6,5136	881806	31,9216	2550156	70,7807	4543029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Quick Sort
Radix Sort
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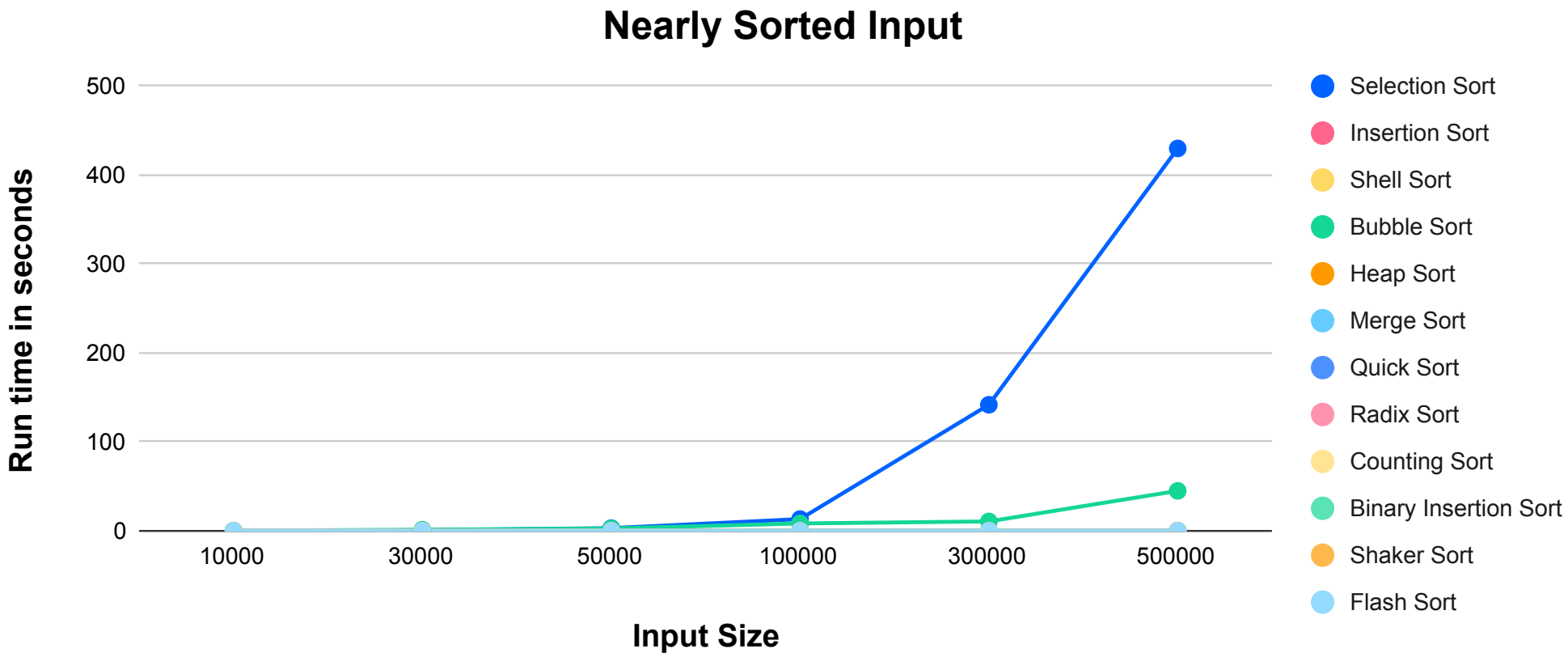
Binary Insertion Sort
Shaker Sort
Flash Sort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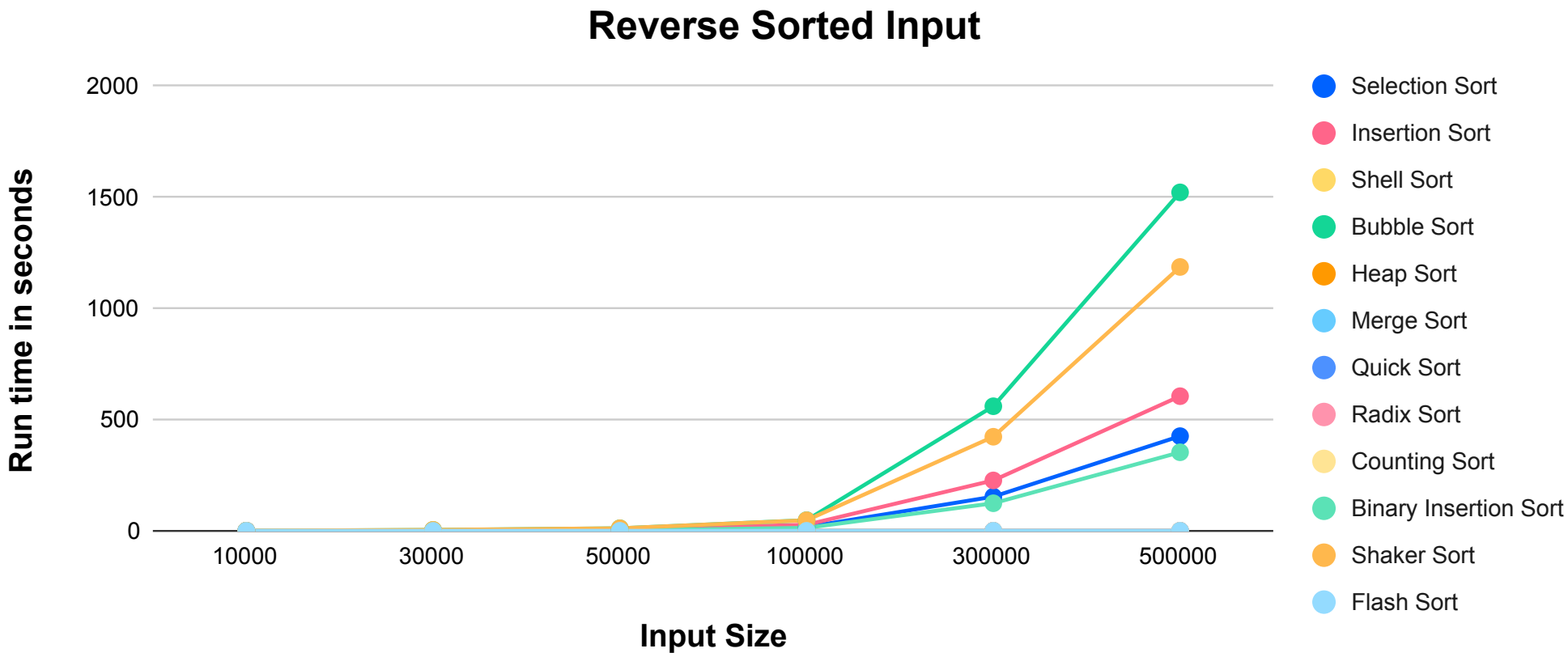
Flash Sort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1000
500
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10000
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Heap Sort
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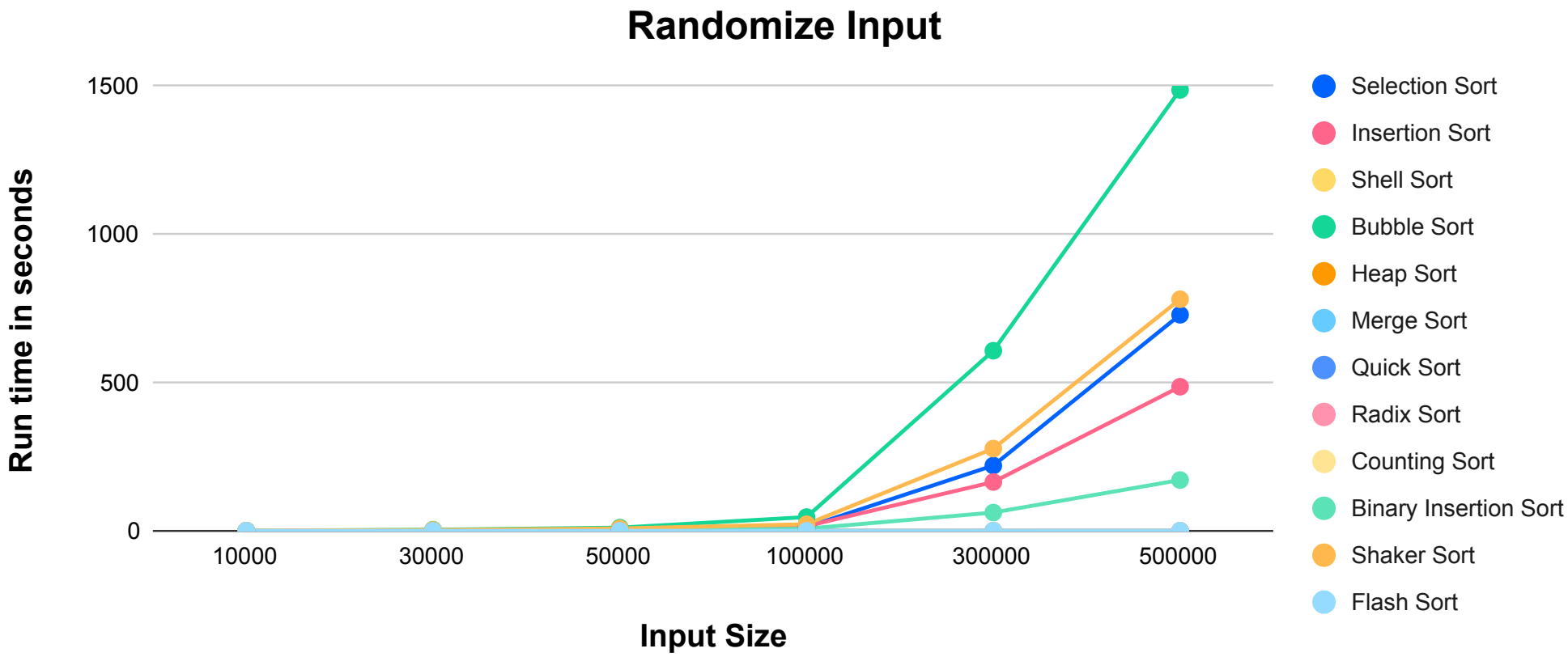
Quick Sort
Radix Sort
Counting Sort
Binary Insertion Sort
Shaker Sort
Flash Sort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Radix Sort
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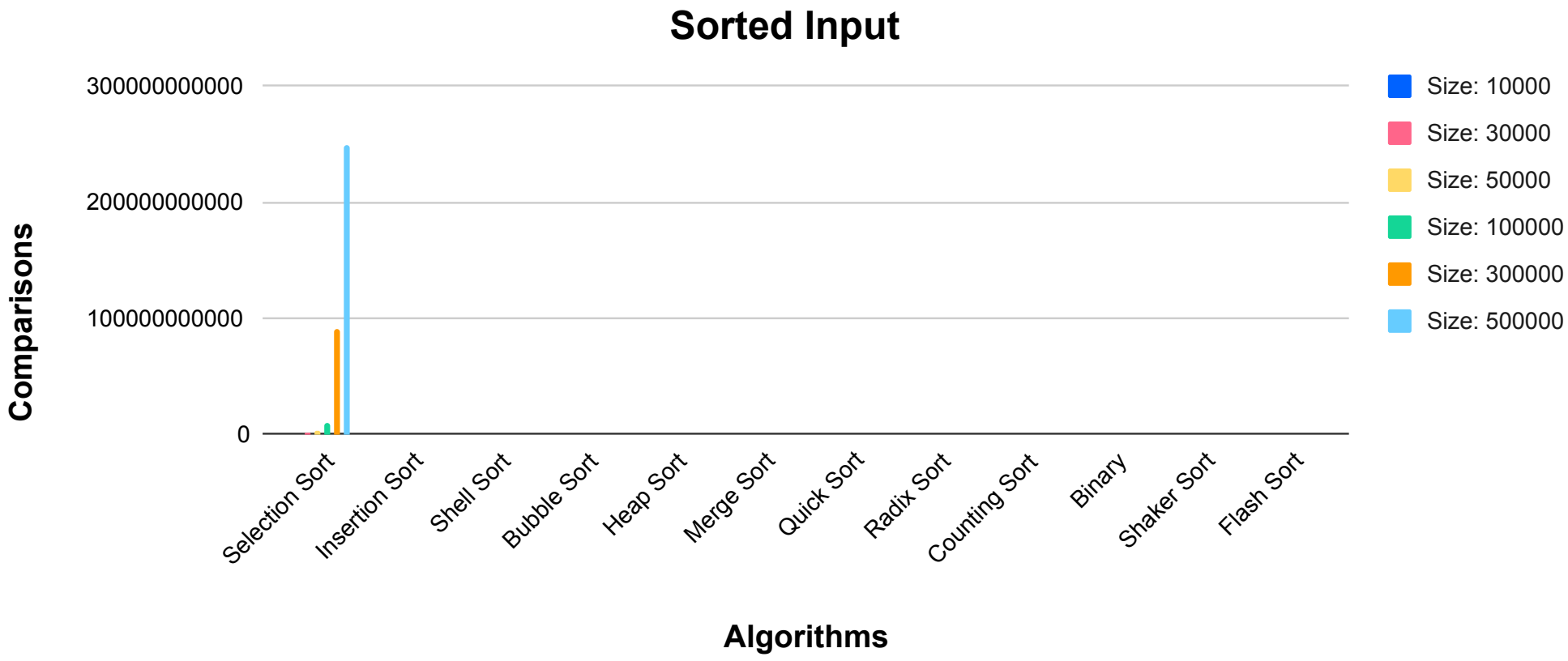
Shaker Sort
Flash Sort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Heap Sort
Merge Sort
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Binary
Shaker Sort
Flash Sort
Algorithms
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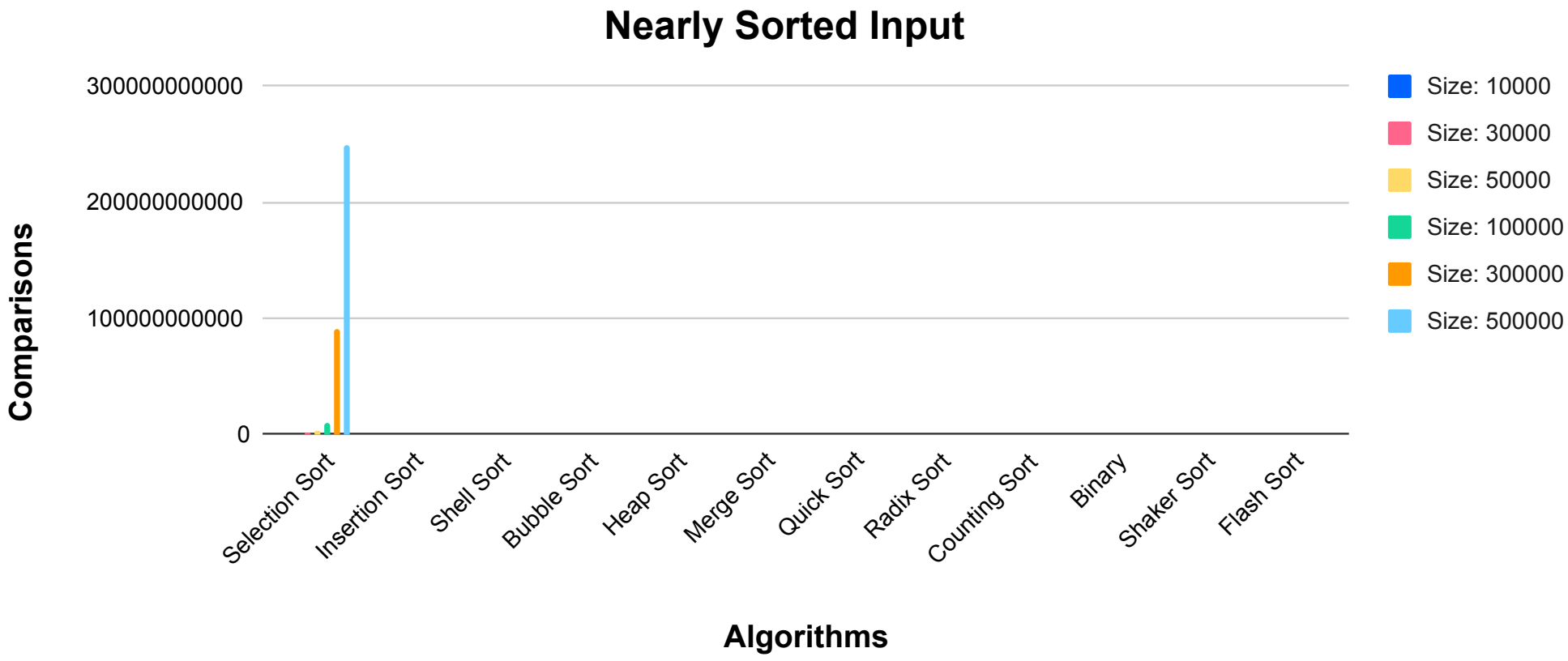
Size: 50000
Size: 100000
Size: 300000
Size: 500000


Nearly Sorted Input
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Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary
Shaker Sort
Flash Sort
Algorithms
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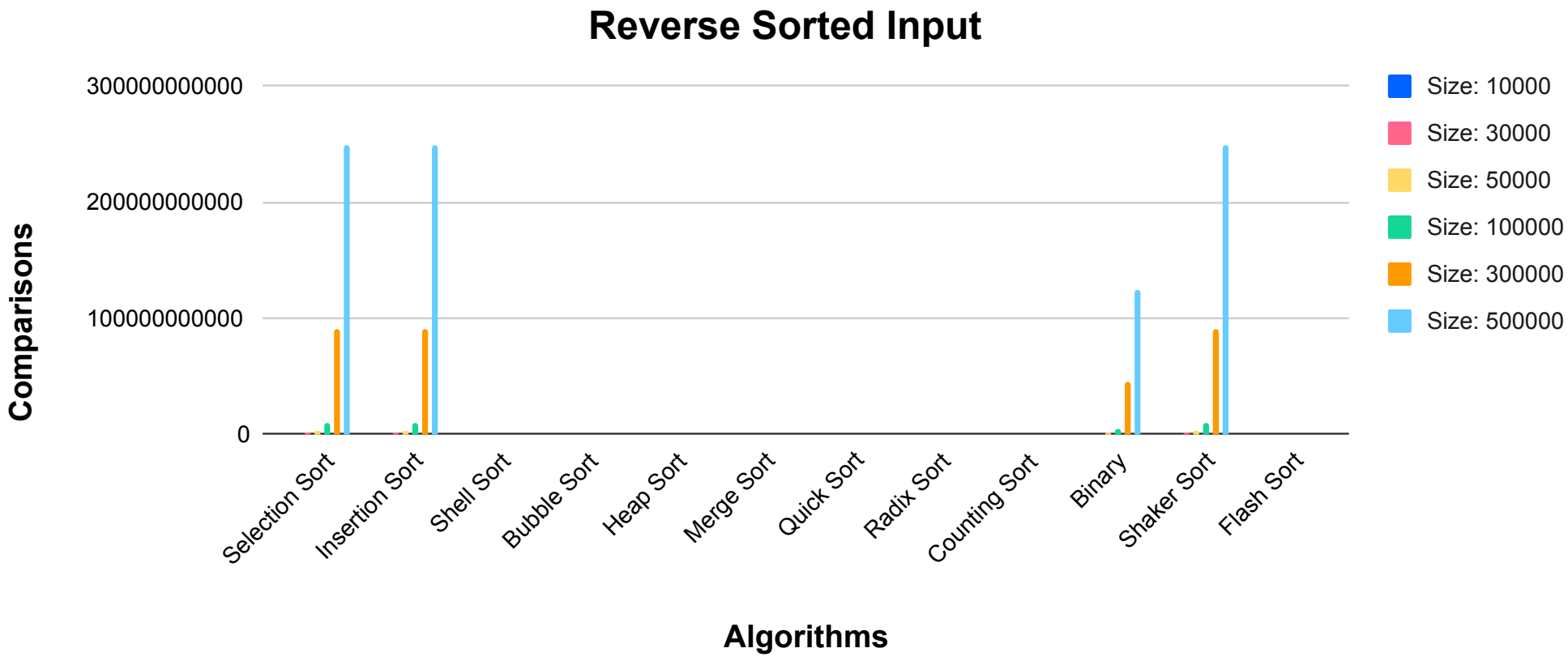
Size: 30000
Size: 50000
Size: 100000
Size: 300000
Size: 500000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Shaker Sort
Flash Sort
Algorithms
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Size: 100000
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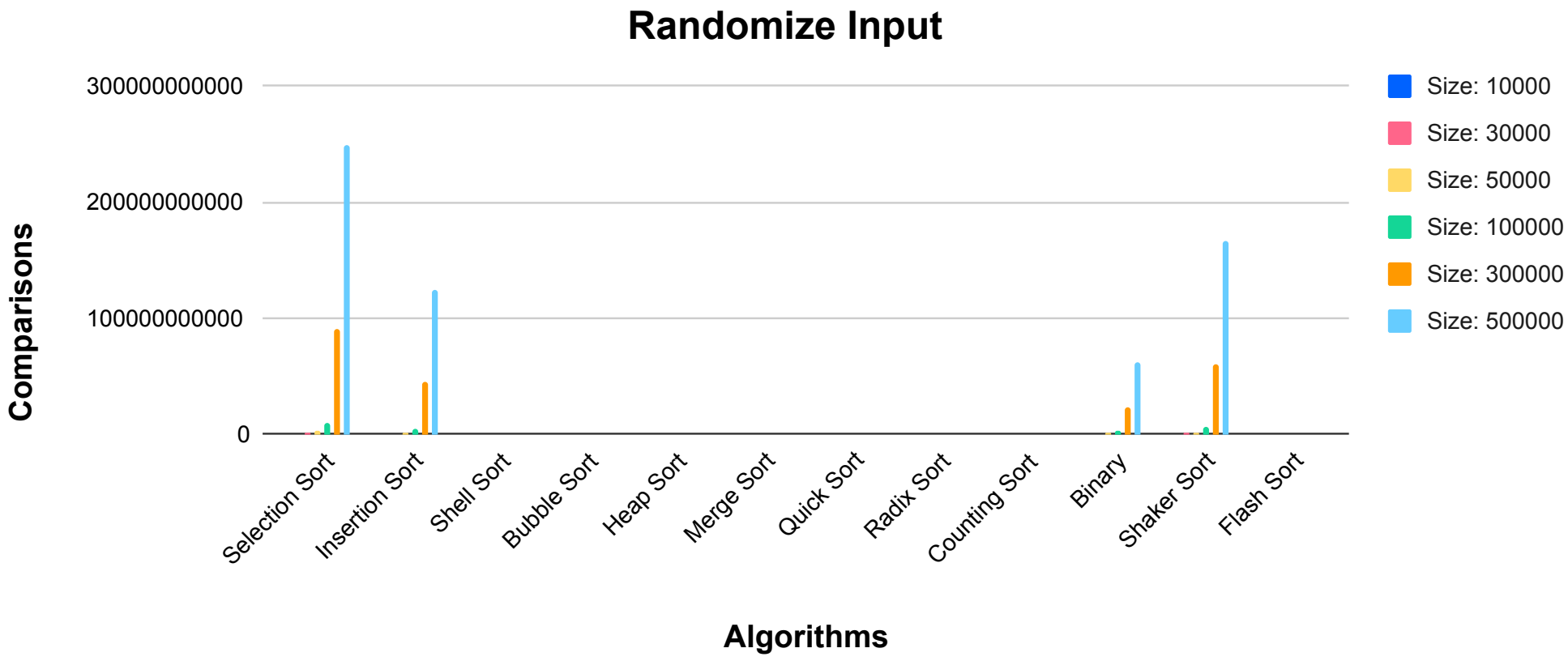
Size: 500000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Comparisons
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0
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
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Radix Sort
Counting Sort
Binary
Shaker Sort
Flash Sort
Algorithms
Size: 10000
Size: 30000
Size: 50000
Size: 100000
Size: 300000
Size: 500000

4 Project Organization and Programming Notes

5 Acknowledgements

6 References

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